

AWS Brain AI: Setup & Deployment Guide

Repository: github.com/vatsala77/aws-realtime-analytics-pipeline

(<https://www.google.com/search?q=https://github.com/vatsala77/aws-realtime-analytics-pipeline>)

Live App: aws-realtime-analytics-pipeline.vercel.app](<https://aws-realtime-analytics-pipeline.vercel.app/>)

1. Ingestion Layer Setup

Step 1.1: IAM Role Configuration (Firehose → S3)

Configure the IAM execution policy and role to grant Amazon Kinesis Data Firehose full write access permissions to your target Amazon S3 data lake bucket.

firehose-to-s3-role [Info](#) [Delete](#)

Allows Firehose to transform and deliver data to your destinations using CloudWatch Logs, Lambda, and S3 on your behalf.

Summary [Edit](#)

Creation date
July 03, 2026, 15:35 (UTC+05:30)

Last activity
 2 hours ago

ARN
 arn:aws:iam::148313549970:role/firehose-to-s3-role

Maximum session duration
1 hour

[Permissions](#) | [Trust relationships](#) | [Tags](#) | [Last Accessed](#) | [Revoke sessions](#)

Permissions policies (1) [Info](#) [Refresh](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

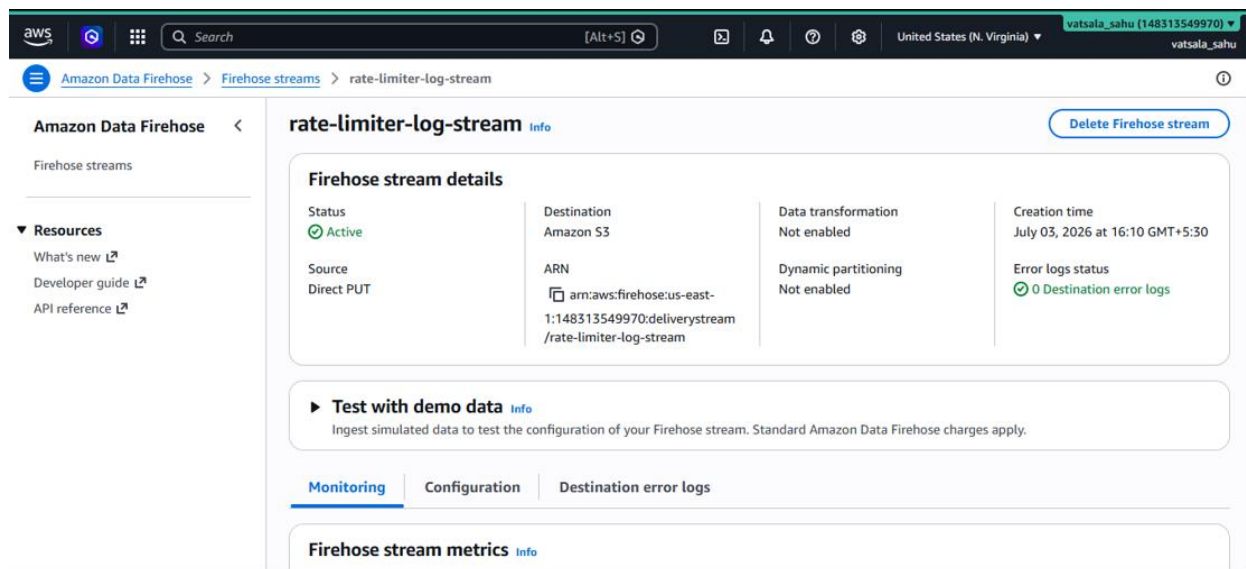
Filter by Type
All types

< 1 >

☐	Policy name ↗	Type	Attached entities
☐	AWSBrainAllLoggerPolicy	Customer inline	0

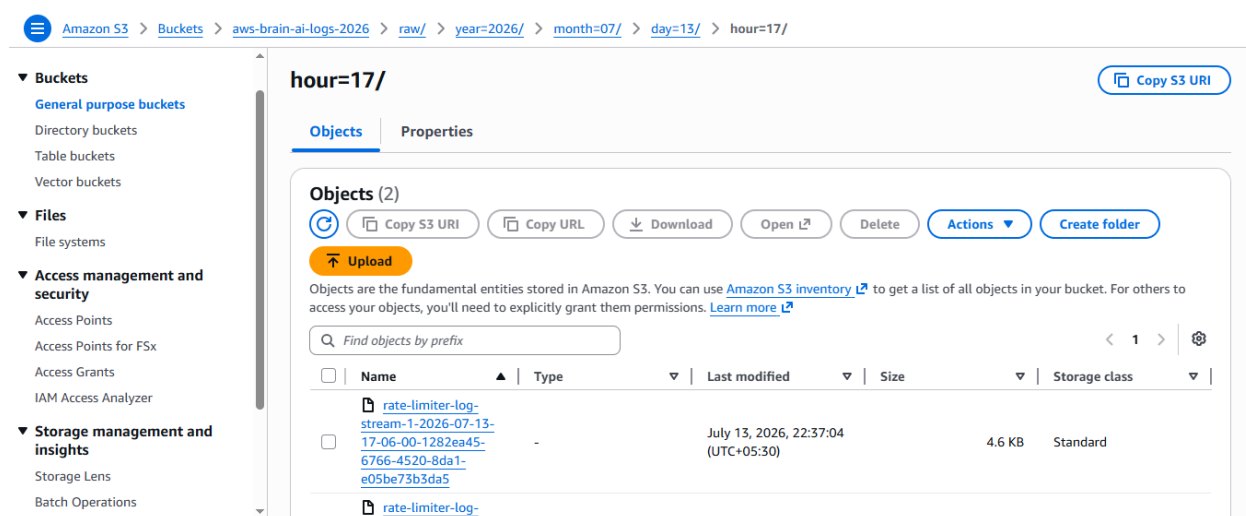
Step 1.2: Kinesis Firehose Delivery Stream Config

Create a new Kinesis delivery stream choosing `Direct PUT` as the source and `Amazon S3` as the destination. Configure the buffer hints explicitly to `60 seconds` or `1 MB`.



Step 1.3: S3 Raw Data Landing with Partition Structure

Verify that Kinesis Firehose is successfully streaming raw JSON logs into your S3 bucket using the automated time-based partition pathing (`YYYY/MM/DD/HH`).



2. Transform & Catalog Layer

Step 2.1: Lambda Function Code & Trigger Configuration

Deploy the Python-based AWS Lambda transformation code and map an active S3
`ObjectCreated` trigger configured exclusively for the `raw/` directory prefix.

The screenshot displays the AWS Lambda console interface for the function 'log-transform-enrich'. The breadcrumb navigation at the top shows 'Lambda > Functions > log-transform-enrich'. The function name 'log-transform-enrich' is prominently displayed at the top left of the main content area. To the right of the name are buttons for 'Throttle', 'Copy ARN', and 'Actions'. Below the function name, there are tabs for 'Diagram' and 'Template', with 'Diagram' being the active tab. The 'Diagram' view shows a visual representation of the function's configuration, including a box for the function name, a box for 'Layers' (currently showing 0 layers), and a box for 'S3' triggers. A '+ Add trigger' button is located below the S3 box. To the right of the diagram, there is a '+ Add destination' button. Further right, there are buttons for 'Export to Infrastructure Composer' and 'Download'. On the far right, there is a sidebar with 'Info' and 'Tutorials' tabs. The 'Tutorials' tab is active, showing a 'Choose a tutorial' section with a link to 'Create a simple web app'. The main content area also includes a 'Function overview' section with 'Info' and 'Export to Infrastructure Composer' buttons. The 'Function overview' section displays the 'Function ARN' as 'arn:aws:lambda:ap-south-1:1483135499:70:function:log-transform-enrich', the 'Description' as '-', and the 'Last modified' time as '3 days ago'. At the bottom of the console, there is a navigation bar with tabs for 'Code', 'Test', 'Monitor', 'Configuration', 'Aliases', and 'Versions', with 'Code' being the active tab.

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

1m 1h UTC timezone Display

Timestamp	Message
2026-07-13T17:07:06.311Z	END RequestId: 650944cd-9eaf-45d3-bd9a-4fe7d63fa9a9
2026-07-13T17:07:06.311Z	REPORT RequestId: 650944cd-9eaf-45d3-bd9a-4fe7d63fa9a9 Duration: 1420.55 ms Billed Duration: 1805 ms Memo...
2026-07-13T17:09:29.521Z	START RequestId: d842ee16-1135-48fa-aa97-9aa8198559cf Version: \$LATEST
2026-07-13T17:09:29.810Z	2026-07-13T17:09:29.810Z d842ee16-1135-48fa-aa97-9aa8198559cf INFO Processing s3://aws-brain-ai-logs-2026...
2026-07-13T17:09:30.230Z	2026-07-13T17:09:30.230Z d842ee16-1135-48fa-aa97-9aa8198559cf INFO Wrote processed file to s3://aws-brain...
2026-07-13T17:09:30.249Z	2026-07-13T17:09:30.249Z d842ee16-1135-48fa-aa97-9aa8198559cf INFO Suspicious count in this batch: 20
2026-07-13T17:09:30.509Z	2026-07-13T17:09:30.509Z d842ee16-1135-48fa-aa97-9aa8198559cf INFO SNS alert published successfully
2026-07-13T17:09:30.531Z	END RequestId: d842ee16-1135-48fa-aa97-9aa8198559cf
2026-07-13T17:09:30.531Z	REPORT RequestId: d842ee16-1135-48fa-aa97-9aa8198559cf Duration: 1008.72 ms Billed Duration: 1009 ms Memo...

No newer events at this moment. Auto retry paused. [Resume](#)

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Step 2.3: AWS Glue Data Catalog Table Schema

Execute the AWS Glue Crawler over the `processed/` bucket directory to automatically discover metadata arrays and generate the structured schema inside the Glue Data Catalog.

Tables

A table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.

Tables (1) Last updated (UTC) July 13, 2026 at 19:37:36 Delete Add tables using crawler Add table

View and manage all available tables.

Choose catalog 148313549970

Choose database rate_limiter_analytics

Filter tables

☐	Name	Database	Location	Classific...	Depreca...	View data	Data quality	Column st...
☐	logs_processed	rate_limiter_ana	s3://aws-brain-a	JSON	-	Table data	View data qualit	View statistics

View and manage the table schema.

Filter schemas

#	Column name	Data type	Partition key	Comment
1	ip	string	-	-
2	method	string	-	-
3	path	string	-	-
4	statuscode	int	-	-
5	algorithm	string	-	-
6	timestamp	string	-	-
7	is_suspicious	boolean	-	-
8	hour_bucket	string	-	-
9	year	string	Partition (0)	-
10	month	string	Partition (1)	-
11	day	string	Partition (2)	-
12	hour	string	Partition (3)	-

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3. Query Layer (Amazon Athena)

Step 3.1: Athena Query Results (All 5 Core Queries)

Launch the Amazon Athena query window to run standard analytical SQL queries targeting specific status codes, event frequencies, or anomaly footprints across the raw tables.

Query 1 : X Query 2 : X Query 3 : X Query 4 : X Query 5 : X Query 6 : X Query 10 : X Query 11 : X Query 12 : X (+) ▼

1 SELECT * FROM logs_processed LIMIT 10;

SQL Ln 1, Col 1

Run again Explain L Cancel Clear Create ▼ Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 52 ms Run time: 689 ms Data scanned: 62.49 KB

Results (10) Copy Download results CSV

Search rows

#	ip	method	path	statuscode	algorithm	timestamp	is_suspicious	hour_bucket	year	month	day
1	1	GET	/api/fixd	200	fixed-window	2026-07-13T05:13:51.118Z	false	2026-07-13T05	2026	07	13
2	1	GET	/api/fixd	200	fixed-window	2026-07-13T05:13:52.071Z	false	2026-07-13T05	2026	07	13
3	1	GET	/api/fixd	200	fixed-window	2026-07-13T05:13:52.777Z	false	2026-07-13T05	2026	07	13

Amazon Athena > Query editor

Editor Recent queries Saved queries Query settings Workgroup: primary

Data

Data source: AwsDataCatalog

Catalog: None

Database: rate_limiter_analytics

Tables and views

Filter tables and views

Tables (1): logs_processed

Views (0)

Query 1: `SELECT ip, method, path, statuscode, timestamp
FROM logs_processed
WHERE is_suspicious = true
ORDER BY timestamp DESC
LIMIT 20;`

SQL: Ln 1, Col 1

Run again Explain L² Cancel Clear Create

Query results Query stats

Completed Time in queue: 58 ms Run time: 811 ms Data scanned: 149.05 KB

Results (20)

Search rows

#	ip	method	path	statuscode	timestamp
1	1	GET	/api/sliding	429	2026-07-13T17:08:47.991Z
2	1	GET	/api/sliding	429	2026-07-13T17:08:47.411Z
3	1	GET	/api/sliding	429	2026-07-13T17:08:46.820Z

4. Anomaly Detection & Alerting

Step 4.1: CloudWatch Logs Showing Suspicious Count & SNS Publish Success

Confirm that when a batch contains a high count of rate-limited `429` requests matching threshold conditions, the Lambda engine outputs a clean execution state logging successful SNS routing.

rate-limiter-anomaly-alerts Edit Delete Publish message

Details

Name: rate-limiter-anomaly-alerts

ARN: arn:aws:sns:ap-south-1:14831354997:0:rate-limiter-anomaly-alerts

Display name: rate_limiter

Type: Standard

Topic owner: 148313549970

Subscriptions Access policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags Integrations

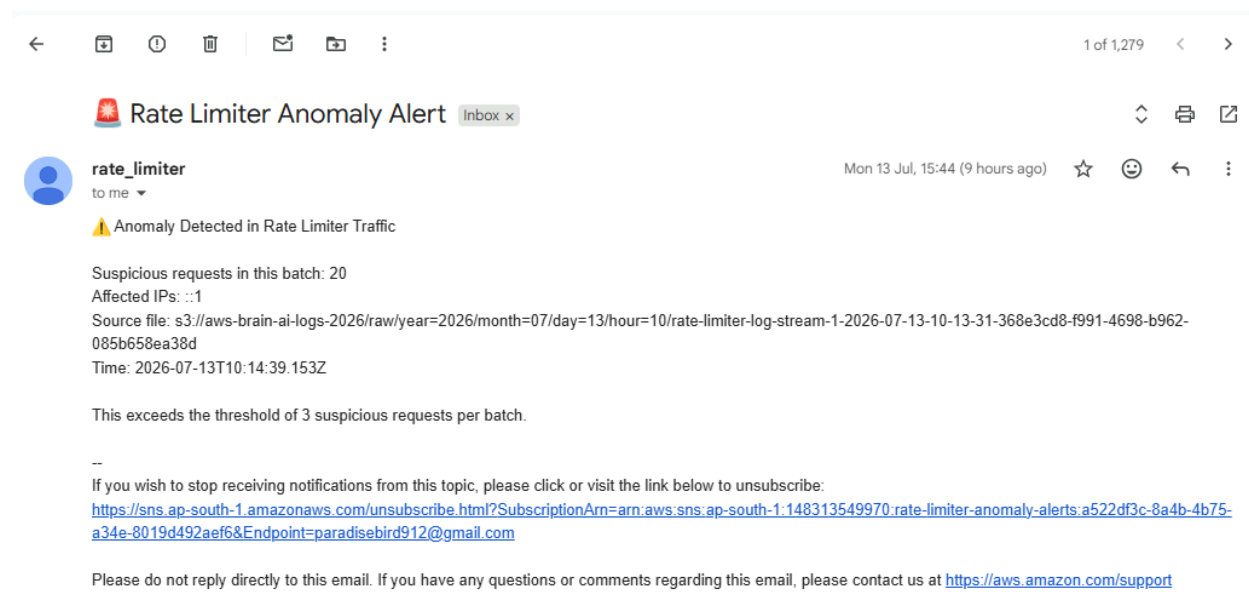
Subscriptions (3) Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol
a522df3c-8a4b-4b75-a34e-8019d492aef6	paradisebird912@gmail.com	Confirmed	EMAIL
ed90fd42-d129-4728-a0ce-097637662119	+919129176768	Confirmed	SMS

Step 4.2: Email/SMS Alert Received

Verify the operational status of the decoupled alert channel by checking the designated delivery mailbox/phone workspace for structural data dispatches.



5. Dashboard & Application Interface

Step 5.1: Live Dashboard with All 5 Charts Populated

Access the live Vercel-hosted React frontend client to verify that your analytical data feeds, graphs, breakdown metrics, and live update tickers load correctly.

Real-Time Traffic Analytics

Live pipeline: Kinesis Firehose → S3 → Lambda → Glue → Athena → Dashboard. The data below reflects actual captured traffic from load-testing sessions — it's not mock data. Use the button below to send new traffic and watch the pipeline update in real time (auto-refreshes every 15s).

 Simulate Traffic Burst

Total Suspicious Events



192

Unique IPs Tracked



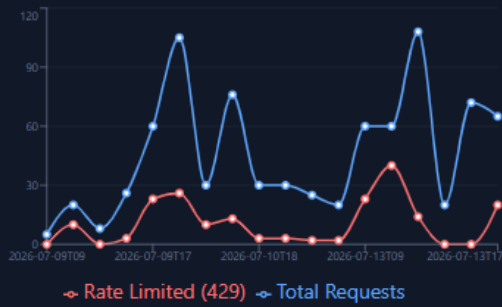
1

Algorithms Monitored



4

Traffic Trend by Hour



Top Offending IPs





Step 5.2: "Simulate Traffic Burst" Interaction

Execute a manual workload stress simulation by triggering the action button and observing the instant analytical feedback flow across tracking tables.

Real-Time Traffic Analytics

Live pipeline: Kinesis Firehose → S3 → Lambda → Glue → Athena → Dashboard. The data below reflects actual captured traffic from load-testing sessions — it's not mock data. Use the button below to send new traffic and watch the pipeline update in real time (auto-refreshes every 15s).

[↻ Simulate Traffic Burst](#)

Traffic sent! Pipeline is processing (Kinesis buffer + Lambda + partition refresh)
— dashboard will reflect new data in ~2 minutes. It'll auto-refresh, no need to reload.